LABORATORY REPORT

Cybersecurity – Module 1

Understanding Basic Security Weaknesses

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Module	Module 01 – Introduction to Cybersecurity

1. Description of the experience

This practical lab aimed to introduce the fundamental concepts of cybersecurity through two main activities: password security analysis and network traffic capture and analysis.

The main tool used was Wireshark, a professional network protocol analyzer that allows inspection of packets exchanged on the network.

1.1 Tools used

- Wireshark (latest version available at wireshark.org)
- Web browser (Firefox / Chrome)
- Operating system: Windows / Linux / macOS

1.2 Procedure

- Analysis of weak passwords and creation of strong passwords
- Launch Wireshark and select the active network interface
- Network traffic generation when browsing multiple websites
- Capture stopped after 2 minutes
- Analysis of captured packets with DNS, HTTP and TLS filters

2. Part 1 – Password Security

2.1 Weak passwords analyzed

The following passwords have been identified as extremely dangerous:

Password	Reason for weakness
`123456`	Trivial numerical sequence, at the top of dictionary lists
`password`	Dictionary word, the most used in the world
`admin`	Default identifier, never secure
`12345678`	Simple extension of 123456, equally predictable
`qwerty`	Keyboard layout, known to all attackers

2.2 Strong passwords created

In accordance with the imposed rules (minimum 12 characters, uppercase letters, lowercase letters, numbers and special characters), the following passwords were generated:

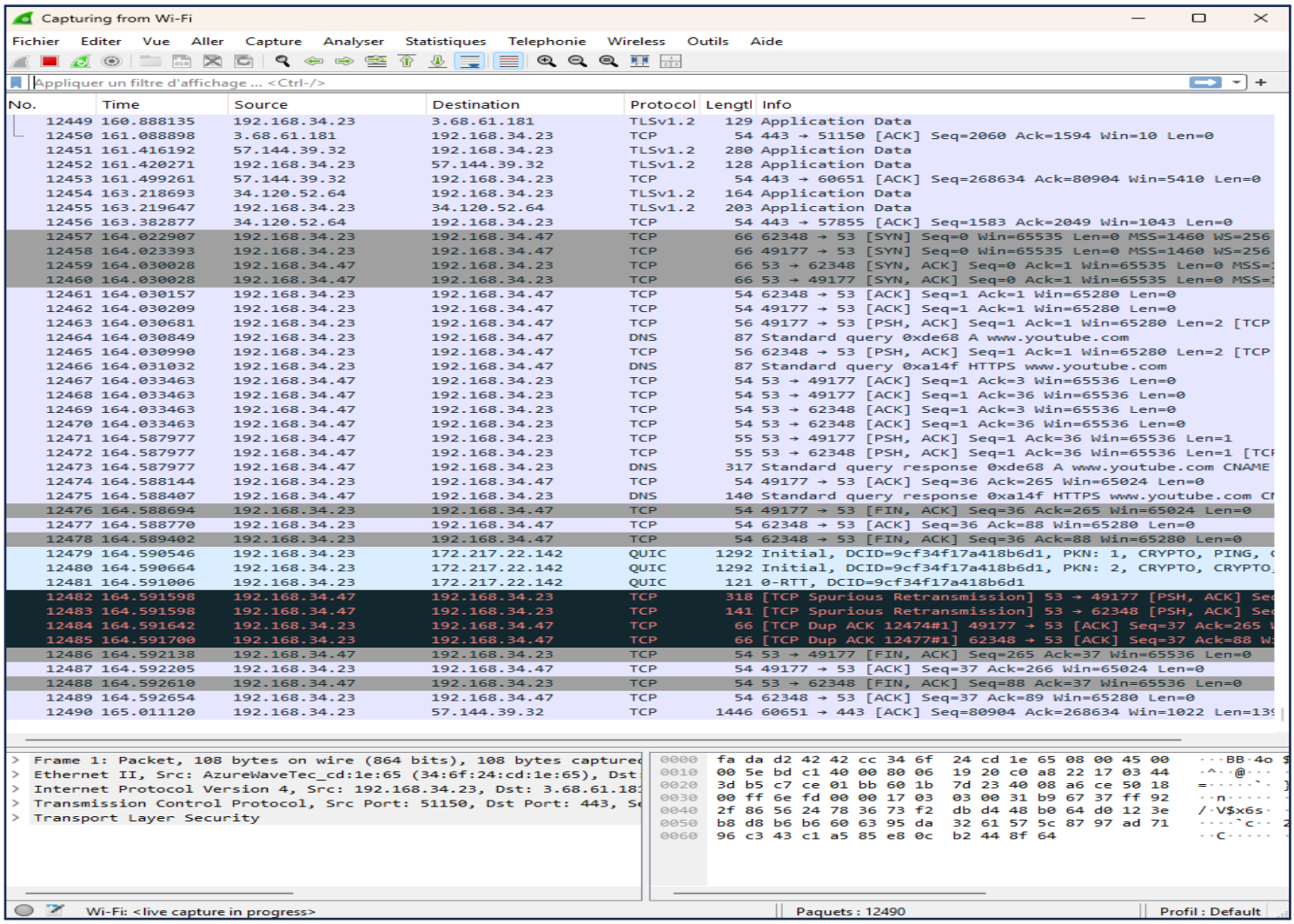
```
1. Tr0pic@lS3cur 2026!
2. F@b13m@Cyber#Lab1
3. W1r3sh@rk&P4ck3ts
```

These passwords adhere to best practices: they are long, complex, dictionary-unguessable, and resistant to brute-force attacks.

3. Wireshark captures

The capture was achieved by browsing Google, Wikipedia , and other websites for approximately 2 minutes. A total of 12,663 packets were captured.

[Screenshot 1 – Wireshark capture in progress (packets scrolling by)]



[Screenshot 2 – DNS filter applied]

*Wi-Fi

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dns

No.	Time	Source	Destination	Protocol	Lengtl	Info
1119	67.176153	192.168.34.47	192.168.34.23	DNS	151	Standard query response 0x9197 HTTPS csp.withgoogle.com SOA ns1.google.com
1156	67.213264	192.168.34.23	192.168.34.47	DNS	86	Standard query 0x5462 HTTPS lh3.google.com
1158	67.213416	192.168.34.23	192.168.34.47	DNS	86	Standard query 0x14e3 A lh3.google.com
1309	67.405737	192.168.34.47	192.168.34.23	DNS	162	Standard query response 0x7b97 HTTPS clientservices.googleapis.com SOA ns1.google.com
1378	67.451903	192.168.34.47	192.168.34.23	DNS	118	Standard query response 0x1a43 A clientservices.googleapis.com A 142.250.129.94
1399	67.482487	192.168.34.47	192.168.34.23	DNS	167	Standard query response 0x5462 HTTPS lh3.google.com CNAME lh2.l.google.com SOA ns1.google.com
1401	67.482873	192.168.34.47	192.168.34.23	DNS	133	Standard query response 0x14e3 A lh3.google.com CNAME lh2.l.google.com A 142.251.142.14
1712	67.683837	192.168.34.23	192.168.34.47	DNS	103	Standard query 0xbdd3 HTTPS content-autofill.googleapis.com
1718	67.684303	192.168.34.23	192.168.34.47	DNS	103	Standard query 0x68bc A content-autofill.googleapis.com
1720	67.684405	192.168.34.23	192.168.34.47	DNS	97	Standard query 0x7fec HTTPS lh3.googleusercontent.com
1724	67.685565	192.168.34.23	192.168.34.47	DNS	97	Standard query 0x05f7 A lh3.googleusercontent.com
1791	67.825473	192.168.34.47	192.168.34.23	DNS	164	Standard query response 0xbdd3 HTTPS content-autofill.googleapis.com SOA ns1.google.com
1792	67.825473	192.168.34.47	192.168.34.23	DNS	312	Standard query response 0x68bc A content-autofill.googleapis.com A 172.217.22.74 A 172.217.16.23…
1803	67.840476	192.168.34.47	192.168.34.23	DNS	208	Standard query response 0x7fec HTTPS lh3.googleusercontent.com CNAME googlehosted.l.googleusercon…
1804	67.840476	192.168.34.47	192.168.34.23	DNS	164	Standard query response 0x05f7 A lh3.googleusercontent.com CNAME googlehosted.l.googleusercontent…
1893	67.942530	192.168.34.23	192.168.34.47	DNS	100	Standard query 0x5415 A ogads-pa.clients6.google.com
1895	67.942585	192.168.34.23	192.168.34.47	DNS	100	Standard query 0x2d81 HTTPS ogads-pa.clients6.google.com
1897	67.942653	192.168.34.23	192.168.34.47	DNS	98	Standard query 0xde6a HTTPS waa-pa.clients6.google.com
1901	67.943251	192.168.34.23	192.168.34.47	DNS	98	Standard query 0xa5bd A waa-pa.clients6.google.com
2078	68.105361	192.168.34.47	192.168.34.23	DNS	161	Standard query response 0x2d81 HTTPS ogads-pa.clients6.google.com SOA ns1.google.com
2079	68.105361	192.168.34.47	192.168.34.23	DNS	117	Standard query response 0x5415 A ogads-pa.clients6.google.com A 142.251.142.10
2080	68.105361	192.168.34.47	192.168.34.23	DNS	159	Standard query response 0xde6a HTTPS waa-pa.clients6.google.com SOA ns1.google.com
2101	68.122364	192.168.34.47	192.168.34.23	DNS	115	Standard query response 0xa5bd A waa-pa.clients6.google.com A 142.251.216.106
2707	70.136053	192.168.34.23	192.168.34.47	DNS	91	Standard query 0x2619 HTTPS clients4.google.com
2709	70.136154	192.168.34.23	192.168.34.47	DNS	91	Standard query 0x9c1f A clients4.google.com
2721	70.294531	192.168.34.47	192.168.34.23	DNS	176	Standard query response 0x2619 HTTPS clients4.google.com CNAME clients.l.google.com SOA ns1.goog…
2722	70.294531	192.168.34.47	192.168.34.23	DNS	142	Standard query response 0x9c1f A clients4.google.com CNAME clients.l.google.com A 216.58.223.206
3021	71.940447	192.168.34.23	192.168.34.47	DNS	86	Standard query 0x0627 A ogs.google.com
3023	71.940495	192.168.34.23	192.168.34.47	DNS	86	Standard query 0x7289 HTTPS ogs.google.com
3052	72.087621	192.168.34.47	192.168.34.23	DNS	168	Standard query response 0x7289 HTTPS ogs.google.com CNAME www3.l.google.com SOA ns1.google.com
3053	72.087621	192.168.34.47	192.168.34.23	DNS	134	Standard query response 0x0627 A ogs.google.com CNAME www3.l.google.com A 172.217.22.46
3145	72.777673	192.168.34.23	192.168.34.47	DNS	87	Standard query 0x9ac1 HTTPS www.gstatic.com
3147	72.777771	192.168.34.23	192.168.34.47	DNS	87	Standard query 0x75dd A www.gstatic.com
3254	72.929887	192.168.34.47	192.168.34.23	DNS	104	Standard query response 0x75dd A www.gstatic.com A 172.217.22.67
3267	72.945245	192.168.34.47	192.168.34.23	DNS	148	Standard query response 0x9ac1 HTTPS www.gstatic.com SOA ns1.google.com
3546	82.042753	192.168.34.23	192.168.34.47	DNS	91	Standard query 0x5528 A clients4.google.com
3548	82.042838	192.168.34.23	192.168.34.47	DNS	91	Standard query 0x543a HTTPS clients4.google.com
3563	82.314572	192.168.34.47	192.168.34.23	DNS	176	Standard query response 0x543a HTTPS clients4.google.com CNAME clients.l.google.com SOA ns1.goog…
3564	82.314572	192.168.34.47	192.168.34.23	DNS	142	Standard query response 0x5528 A clients4.google.com CNAME clients.l.google.com A 216.58.223.206
3656	86.263183	192.168.34.23	192.168.34.47	DNS	89	Standard query 0x7210 HTTPS www.wikipedia.org
3658	86.263241	192.168.34.23	192.168.34.47	DNS	89	Standard query 0x6422 A www.wikipedia.org
3667	86.857687	192.168.34.47	192.168.34.23	DNS	173	Standard query response 0x7210 HTTPS www.wikipedia.org CNAME dyna.wikimedia.org SOA ns0.wikimedi…
3668	86.857687	192.168.34.47	192.168.34.23	DNS	138	Standard query response 0x6422 A www.wikipedia.org CNAME dyna.wikimedia.org A 185.15.58.224
3688	87.229118	192.168.34.23	192.168.34.47	DNS	92	Standard query 0x3c14 A storeedgefd.dsx.mp.microsoft.com
3689	87.391694	192.168.34.47	192.168.34.23	DNS	210	Standard query response 0x3c14 A storeedgefd.dsx.mp.microsoft.com CNAME storeedgefd.dsx.mp.micro…
3774	88.957184	192.168.34.23	192.168.34.47	DNS	92	Standard query 0xe1e4 HTTPS upload.wikimedia.org
3776	88.957293	192.168.34.23	192.168.34.47	DNS	92	Standard query 0x9220 A upload.wikimedia.org
3812	89.187110	192.168.34.47	192.168.34.23	DNS	109	Standard query response 0x9220 A upload.wikimedia.org A 185.15.58.240
3813	89.187110	192.168.34.47	192.168.34.23	DNS	144	Standard query response 0xe1e4 HTTPS upload.wikimedia.org SOA ns0.wikimedia.org
3853	89.320777	192.168.34.23	192.168.34.47	DNS	88	Standard query 0x7196 HTTPS fr.wikipedia.org
3855	89.320842	192.168.34.23	192.168.34.47	DNS	88	Standard query 0xb5b9 A fr.wikipedia.org
3864	89.324324	192.168.34.23	192.168.34.47	DNS	103	Standard query 0xe484 A content-autofill.googleapis.com
3866	89.324410	192.168.34.23	192.168.34.47	DNS	103	Standard query 0xcc34 HTTPS content-autofill.googleapis.com

Packet (115 bytes) Reassembled TCP (62 bytes)

Domain Name System: Protocol

Paquets : 12663 · Affichés : 180 (1.4%) · Perdus : 0 (0.0%)

Profil : Default

[Screenshot 3 – HTTP Filter (no results)]

*Wi-Fi

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http

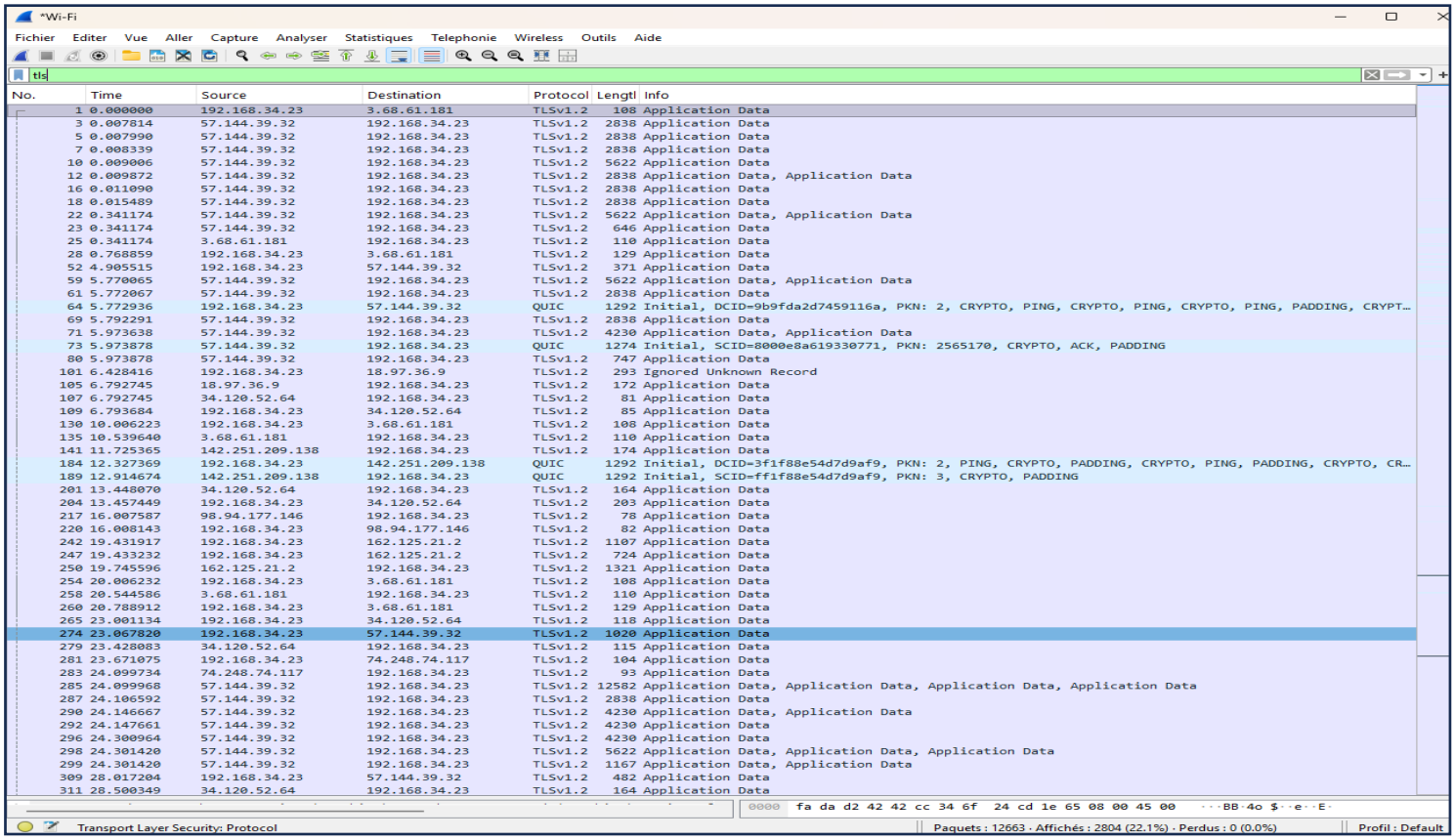
No.	Time	Source	Destination	Protocol	Lengtl	Info

Hypertext Transfer Protocol: Protocol

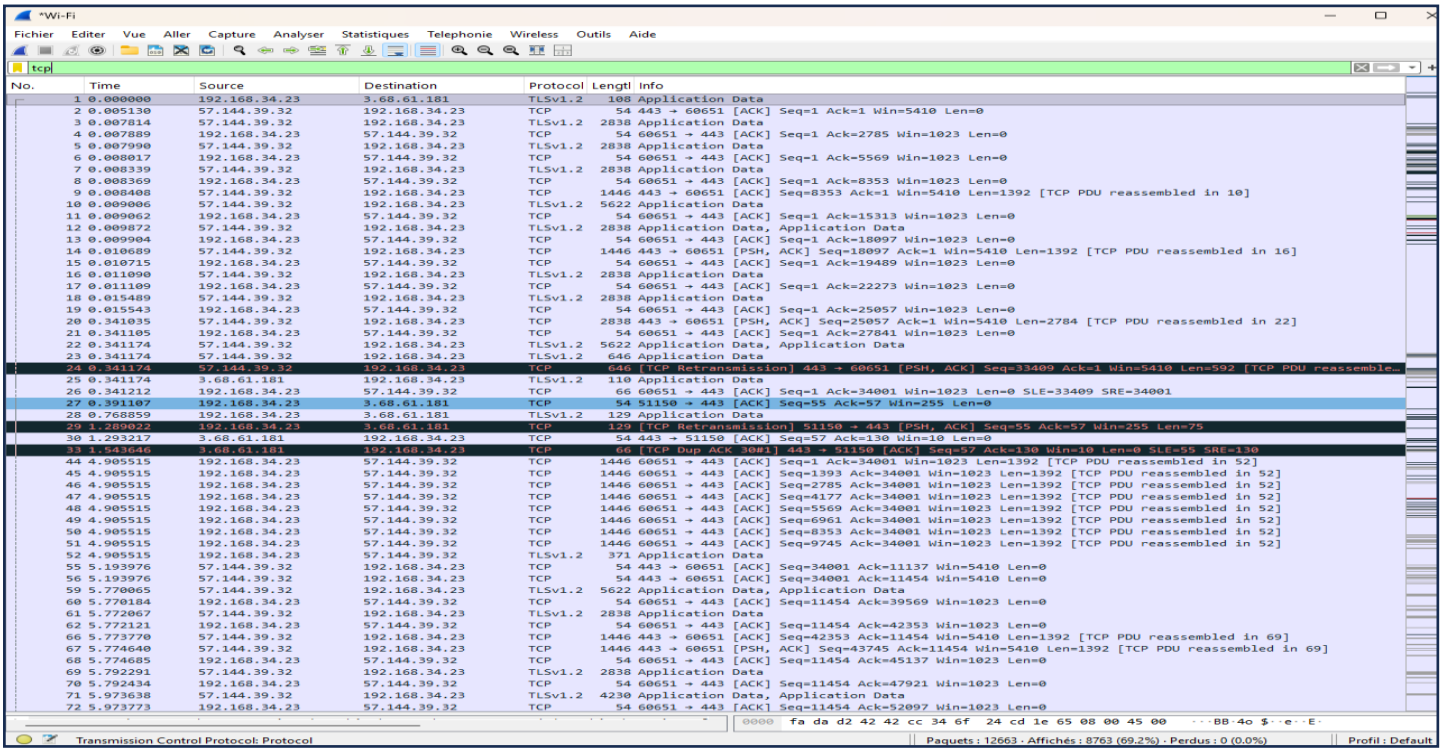
Paquets : 12663 · Affichés : 0 (0.0%) · Perdus : 0 (0.0%)

Profil : Default

[INSERT HERE: Screenshot 4 – TLS filter applied]



[Screenshot 4 – TCP filter applied]



4. Protocols observed

During this experiment, the following network protocols were identified:

Protocol	Full name	Role
DNS	Domain Name System	Domain name resolution to IP addresses
TLS	Transport Layer Security	Encryption of HTTPS communications
TCP	Transmission Control Protocol	Reliable data transfer between machines

Note: No unencrypted HTTP packets were observed, which confirms that modern sites exclusively use HTTPS.

5. Answers to the analysis questions

Question 1	12,663 packets were captured during the approximately 2-minute session.

Question 2	Three protocols observed: DNS (domain name resolution), TLS (encrypted HTTPS communications), TCP (reliable data transport).

Question 3	Encryption is essential to protect sensitive data (passwords, personal information, banking data) against interception by a third party on the network.

Question 4	On unencrypted HTTP traffic, an attacker can retrieve visited URLs, usernames and passwords submitted via forms, as well as any personal data transmitted in plain text.

6. Safety Lessons Learned

6.1 About passwords

- Weak passwords can be guessed in seconds by automated tools (dictionary attacks, brute force).
- A strong password should be at least 12 characters long with a combination of character types.
- It is recommended to use a password manager for each service.

6.2 On network traffic

- Any network traffic can be captured by anyone on the same network.
- Unencrypted HTTP communications fully expose data to malicious third parties.

- The TLS/HTTPS protocol guarantees the confidentiality of communications over the Internet.
- The total absence of HTTP packets during our browsing demonstrates that modern sites have generalized HTTPS, which is a good practice.

6.3 Best practices to remember

- Always use complex and unique passwords for each service.
- Check for the presence of the padlock (HTTPS) before entering sensitive information.
- WiFi networks for sensitive transactions.
- Use a VPN to encrypt all outgoing traffic.

7. Conclusion

This lab provided a practical understanding of two fundamental pillars of cybersecurity: password management and network communication security. The capture of 12,663 packets in just two minutes illustrates the omnipresence of network traffic and the potential risks that arise without adequate protective measures.

The complete absence of unencrypted HTTP traffic during our tests confirms that the industry has largely adopted TLS encryption as the standard. However, this protection does not eliminate the need for constant vigilance regarding the strength of passwords used and the networks to which one connects.

In short, cybersecurity relies on a combination of good technical practices (encryption, strong authentication) and behavioral practices (vigilance, digital hygiene) that every digital professional must master.